

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishmentsof the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts andvalue of funding unique partnerships to support innovative science addressing today’s food and agriculture challenges. If a question is not applicable to the project, please type “Nothing to report.”

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Project Title	Project Ella
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Matching Funder(s)	In Ovo B.V.
Final Report Date	01/10/2025
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Section I

Please explain below how the in-ovo sexing technology meets the Egg-Tech Prize Evaluation Criteria. Respond to all six prompts and upload the completed document as a PDF.

1. Day and potential to utilize male eggs (<250 words)

Minimum: Functions on or before day 11 of incubation, with day-0 counted as the day the eggs go into the setter and the following calendar day being day-1. Preference for solutions with reduced incubation time with pre-incubation most preferred. Protocols involving short periods of incubation during egg storage (SPIDES) will be considered pre-incubation and given preference. Preference will be given to technologies that enable the use of male eggs in other applications. *Please explain below how your in-ovo sexing technology meets this evaluation criteria.*

Our gendering solution is effective both on day 11 of incubation and earlier. In hatcheries using the Ella system, gender screening is typically performed between days 9 and 11 of incubation. This timing ensures compliance with German regulations, which require screening before day 13, while also allowing testing across multiple incubation days to support the production of large flocks on the same hatch date.

We have demonstrated differential biomarker expression in allantoic fluid as early as day 6 and are actively working to shift the testing window to day 6—a goal we aim to achieve in the coming year. Our mass spectrometers can screen multiple biomarkers simultaneously without reducing throughput. We are also adding an additional biomarker to further enhance the accuracy of our gendering solution and enable earlier screening.

The ability to perform screenings as early as day 6 will significantly reduce the incubation time for male eggs, freeing up valuable setter capacity. Our current workflow, however, already ensures that 100% of the eggs undergoing gender screening are utilized. Unfertilized eggs, early embryo losses, and male eggs are immediately separated from female eggs during the sorting process. These non-viable eggs are repurposed and valorized by Symrise for the production of sustainable egg proteins used in pet food. Symrise operates in various markets, including the US.

Additionally, our gendering solution is compatible with batches that have undergone SPIDES treatment, with no negative impact on screening performance.

2. Accuracy (<250 words)

Minimum: 98% accuracy. Preference will be given to technologies that work with all chicken breeds/colors commonly used in commercial production. *Please explain below how your in-ovo sexing technology meets this evaluation criteria.*

Our technology works across all chicken breeds, regardless of color, as it targets a biomarker directly linked to gender that is universally expressed in all breeds.

Over the past year, significant modifications to Ella have dramatically improved sample quality, increasing gendering accuracy from 96% to 98.5% while enhancing overall process reliability. A key factor in this improvement has been the full automation of sample handling and downstream egg sorting. This automation not only accelerates the sorting process but, more importantly, eliminates the risk of human error affecting gendering accuracy.

Additionally, we have refined our gendering algorithm to account for variations in sample quality caused by lower egg quality, slight differences in egg size, and varying incubation times. These advancements have collectively resulted in a substantial increase in gendering accuracy.

At hatchery "Het Anker," we now operate at a stable 98.5% gendering accuracy. We have already produced batches exceeding 99% accuracy and anticipate that with the integration of a second biomarker, we will consistently achieve over 99% gendering accuracy this year.

3. Economic Feasibility (<1,000 words)

Score for this criterion will consider economic feasibility based on a cost-benefit analysis and business plan that should include:

Direct costs --

- Capital costs incurred by technology developer, per hatchery
- Capital investment for equipment/structure modification by hatchery
- Predicted annual maintenance costs
- Predicted annual consumables costs
- Predicted personnel training and labor requirements (hours)

Indirect costs --

- Expected utilities requirements of technology

Potential revenue models --

- Lease, subscription, sales, other.
- Other revenue streams for developer
- Predicted revenues gained for hatchery in diverting eggs, energy savings, labor, cost-savings from not feeding male chicks (depending on country), etc. Please explain below how your in-ovo sexing technology meets this evaluation criteria.

Table1 shows a complete overview of all costs, costs savings and spin-off revenues linked to running Ella. All these three are also calculated per female chick. The table consists of four sections, each explained here:

1. Current Ella input figures. This section shows an Ella line based on an average sized hatchery. This means that it does 15,000 tests per hour on our current performance. This line produces -based on a runtime of 2,000 hours per year and our current test ratio of 2.45- 12.2M female chicks per year.
2. Costs involved in placing and running Ella. Capital costs are \$14M, maintenance p/yr \$400k and \$100k is needed for structure modification (explained under 4. Throughput and Physical size).
3. Assumptions costs & revenue streams. There is a section with assumptions on costs and revenue streams for the US market. We assume in this model that male chicks that hatch as a result of sexing errors will be reared at a cost of \$3.80 per chick. We produce currently with an egg ratio of 2.7 (test ratio of 2.45 + 10% unfertilized eggs) which is 0.2 eggs more than a hatchery uses without in ovo sexing. The costs of these 0.2 extra hatching eggs per female chicks are in the model.
4. Cost calculation per female chick. This section gives a division of the costs per female chick between In Ovo and the Hatchery. These are the following for In Ovo:
 - Capital costs machine. In Ovo pays for the machine and leases it to the hatchery. The capital costs per female chick are calculated based on a 3 year payback period of the machine and a production of 12.2 mln female per year.
 - Consumables. \$0.04 per test and 2.45 tests per female result in costs of \$0.10 per female chick.
 - Maintenance: \$400k per year results in \$0.03 per female chick.

The costs, costs savings and spin-off revenues for the hatchery are the following:

- Personnel. the personnel required to run Ella is three people, two to take and bring trolleys from and to the incubator and one person to monitor the process. Assuming a labour cost p/hr of \$30 and 2,000 runtime hours, this comes to \$0.01 per female chick. Personnel is usually employed by the hatchery, but can also be provided by In Ovo. In both cases, the personnel costs are paid by the hatchery.
- Male chick rearing. To ensure a fully cull-free production, male chicks resulting from sexing errors are reared. At the current 98.5% accuracy, this cost amounts to \$0.06.
- Extra eggs. We currently need 2.70 eggs to hatch a first grade female chick. Assuming a hatchery with regular practice uses 2.50 eggs, these 0.2 extra eggs per female chick cost \$0.03.
- Energy. Based on 2,000 hours runtime, Ella uses about 50,000 kWh in a year. At a price of \$0.20 per kWh, this amounts to \$0.001 per chick.
- Structure modification. Calculating this over 1 year production of 12.2M chicks, this comes to \$0.004 per female. Spreading this over more years lowers this cost on a per female chick basis.
- Cost saving manual sexing. Manual sexing of a chick costs about \$0.02. With



our current accuracy of 98.5%, nearly all males are taken out of the incubation process and the personnel only have to sex 50% of the chicks they had to sex before, which amounts to a cost reduction of \$0.01.

- Costs saving setters energy. It costs about \$0.03 to hatch an egg. As we take the unfertilized eggs (~10%) and males (45%) out between day 9-11 of the 18 day incubation process, this leads to a 24% reduction in energy usage during incubation. In the hatching stage the reduction is 50%. As hatching is only 3 days, overall this comes to a reduction of 28% in energy usage, resulting in \$0.008 reduction per female chick.
- Revenue repurposing male eggs. These male egg are in Europe used as a pet feed ingredient and picked up, generating a revenue of about \$0.005.
- Cost saving – switch to in ovo vaccination. In ovo sexing makes it possible to apply in ovo vaccination on day 18, as only females are transferred and now vaccine is used on males. This has a spin-off reduction of personnel costs that is estimated to be on the same level as the personnel needed for production (\$0.02 per female chick).

Total costs:

- Current system: The total costs for producing a completely cull-free chick come to **\$0.39**, of which **\$0.32** is borne by In Ovo and **\$0.07** by the hatchery.
- Accuracy >99%, 2025: An important remark to be made is that when we reach a level of accuracy that is consistently above 99% in 2025, the expectation is that the max. 1% of males will be reared. This will remove the hatchery costs for male chick rearing \$0.06 and the costs saving on manual sexing will double to (\$0.02). At that time total production costs will be reduced to **\$0.32** of which **\$0.31** are borne by In Ovo and **\$0.01** by the hatchery.

Revenues and margins In Ovo and hatchery

These cost levels allow for a very healthy margin to be made on cull-free chicks (Table 2). As these chicks meet higher animal welfare standards than regular chicks, this adds value to the eggs they produce. In Germany this added value of the eggs translates to a price level of in ovo sexing at chick level of \$2.90, leaving a margin of \$2.51. The current market price as communicated by NestFresh in this [Agfunder](#) article stated that the added value in the US will be around \$1.50 leaving **a margin of \$0.92 going up to \$0.99** when the accuracy is consistently above 99% in 2025.

Going from a market price of \$1.50 In Ovo will expect to charge a price per chick of \$0.99 to hatcheries leaving a margin of \$0.48 for In Ovo and \$0.51 for the hatchery. Based on the output of 12.2 mln chicks p/yr, the hatchery will make \$18.3 mln revenue with \$6.2 mln margin. In Ovo will make \$12.1 mln revenue and \$6.2 mln margin (Table 3).

Table 1

1. Current Ella input figures			
Throughput (tests/hour)	15,000		
Test ratio	2.45		
Egg ratio	2.70		
Accuracy	98.5%		
Runtime per year (hours)	2,000		
Females/year	12,244,898		
2. Costs involved in placing and running Ella		Per female chick produced	
Capital costs machine	\$ 14,000,000	\$	0.38
Payback period machine (yrs)	3		
Personnel	3	\$	0.01
Consumables p/test	\$ 0.04	\$	0.10
Maintenance costs p/y	\$ 400,000	\$	0.03
Energy consumption (kwh/yr)	50,000		
Structure modification (1 time)	\$ 100,000	\$	0.008
3. Assumptions costs & revenue streams		Per female chick produced	
Rearing costs males (per male chick)	\$ 3.80	\$	0.06
Labour (per hour)	\$ 30.00	\$	0.01
Costs extra hatching eggs (per hatching egg)	\$ 0.17	\$	0.03
Energy costs Ella (per kWh)	\$ 0.20	\$	0.001
Manual sexing (per sexed chick)	\$ 0.02	\$	(0.01)
Revenue repurposing male eggs (per male egg)	\$ (0.005)	\$	(0.005)
Energy costs incubation (per egg)	\$ 0.03	\$	(0.008)
Cost saving - switch to in ovo vacc (per egg)	\$ (0.02)	\$	(0.02)
4. Costs and savings - split In Ovo and Hatchery		In Ovo costs	Hatchery costs
Capital costs machine	\$	0.38	
Consumables	\$	0.10	
Maintenance	\$	0.03	
Labour			\$ 0.01
Rearing costs males			\$ 0.06
Extra hatching eggs			\$ 0.03
Energy consumption			\$ 0.001
Structure modification			\$ 0.008
Manual sexing			\$ (0.01)
Costs saving setters energy			\$ (0.008)
Revenue repurposing male eggs			\$ (0.005)
Cost saving - switch to in ovo vacc			\$ (0.02)
Total costs per female chick	\$	0.51	\$ 0.07
			\$ 0.58

Table 2

Margin p/chick	Germany 2024	US (projection)
Added value of cull free chick	\$ 2.90	\$ 1.50
Total costs per female chick	\$ 0.58	\$ 0.58
Margin per cull free female chick	\$ 2.32	\$ 0.92

Table 3

Revenues and margins hatchery and In Ovo		
Market price in ovo sexing	\$ 1.50	
Number of chicks per year	12,244,898	
Total revenue hatchery	\$ 18,367,347	
Price In Ovo p/chick		\$ 0.99
Revenue In Ovo p/yr		\$ 12,122,449
Costs In Ovo p/chick		\$ 0.51
Margin In Ovo p/yr		\$ 6,204,000
Margin Hatchery p/yr	\$ 6,244,898	

4.Throughput and Physical Size (<250 words)

Potential for least 15,000 eggs per hour. If multiple units will be used in combination to achieve the desired throughput, only one demonstration unit will be required but all units needed to meet the desired throughput must fit into existing hatchery structures, with reasonable and appropriate modifications. *Please explain below how your in-ovo sexing technology meets this evaluation criteria.*

Our current Ella system at hatchery Het Anker has a throughput of 10,400 eggs per hour and includes the following components:

- 2 samplers
- 4 Echo MS systems
- Trolley unloader and loader at the start and end of the line
- Heartbeat candler, that removes unfertilized eggs and early deaths, with backfilling of fertile eggs before entering the sampler
- Sorter with backfilling of female eggs at the end of the line

The setup is modular, meaning that adding one additional sampler and one Echo MS system would increase the capacity to 15,000 eggs per hour. The peripherals for candling, (un)loading, sorting, and backfilling are already designed to handle 15,000 eggs per hour. The total footprint of this upgraded configuration is 2,690 sq ft.

Our next-generation Ella machine is currently under development and is expected to be ready by the end of 2025. It features a redesigned sampler and a more compact configuration that supports higher throughput within a smaller footprint. In a setup

designed to process 15,000 eggs per hour, the footprint would be reduced to 2,150 sq ft. In addition to the required floor space, hatcheries need to install the following:

- Air: Clean air inlet and waste air outlet
- Power supply connections:
 - 1x 400VAC 32A
 - 6x 400VAC 16A
 - 6x 230VAC 16A
- Compressed air: 7 bar (25 m³ per hour)
- Internet connection
- Water supply: ¾" water pipe connection, water temperature between 10°C and 25°C
- Water drainage: minimum 40 mm pipe

5. Hatchability (<250 words)

Minimum: Does not reduce hatching rate > 1.5% from baseline. *Please explain below how your in-ovo sexing technology meets this evaluation criteria.*

Ella currently operates with an additional hatch impact of just 0.5%, well below the 1.5% benchmark compared to the baseline (see figure 1). The primary factors contributing to hatchability impact in our system are cross-contamination and physical damage.

To minimize contamination, incoming eggs are sanitized with UVC light. During the process, the egg punches are heated for disinfection, and the sampling needles are cleaned after each sample collection. We adhere to strict protocols to monitor hygiene levels for both the machine and the Ella room, ensuring that all cleaning procedures meet rigorous standards. By taking these steps ourselves to prevent cross-contamination of eggs, no additional hygiene measures are required for the hatchery. To prevent eggs from cooling during sampling, we also minimize the time they spend outside the incubator.

Throughout the sampling and sorting process, we continuously monitor shock levels and eggshell temperatures using egg sensors. This data has led to multiple design improvements in the Ella sampler to further reduce shocks and temperature fluctuations. Additionally, increased automation has reduced the number of manual handling steps, contributing to a consistently lower hatch impact.

The hatchery where our two gendering machines are installed reports an annual average hatchability of 93% without using Ella, with a 7% unhatched rate. We benchmark our performance against this baseline. For example, over the past four weeks (weeks 42–46), the average hatchability of gendered eggs was 92.5%, with a standard deviation of 0.6%. This demonstrates that our system performs consistently, with only a -0.5% difference compared to the hatchery's annual average.

Hatchability

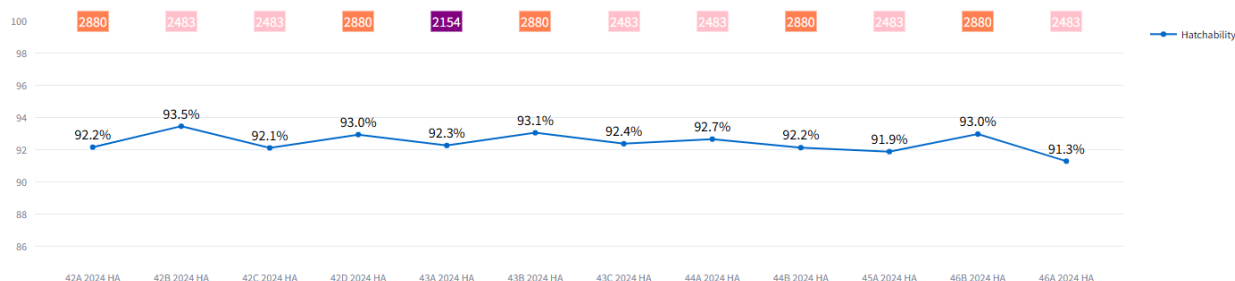


Figure 1. Hatchability after in ovo sexing with Ella in week 42-46, 2024

6. Speed of test results (<250 words)

< 30 min for test results if eggs are tested during incubation (allowable time for removal, testing and return to incubator). If eggs are tested prior to incubation (with or without SPIDES), results must be available within 48 hours of testing. Accurate tracking and identification of eggs must be demonstrated. *Please explain below how your in-ovo sexing technology meets this evaluation criteria.*

Ella is an inline system that screens eggs during incubation, completing the entire process—from removing unsexed eggs from the trolley to returning female eggs to the incubator—in just 25 minutes.

The screening occurs between 9 and 11 days after the start of incubation. After automated unloading, the eggs undergo a candling process to identify and remove unfertilized eggs and early deaths. Each remaining egg is sampled using the Ella sampler, depositing a sample into a corresponding position on a sample well plate. Each egg is assigned a unique virtual ID linked to its sample location within the well plate.

Following sample preparation, the well plates are analyzed using ECHO-MS technology. The analysis quantifies the results and determines the gender of all 380 samples simultaneously. Thanks to the virtual IDs, the gendering results are linked back to the individual eggs.

This data is then sent to the automated sorting system, which separates the eggs based on their determined gender. Trays are filled exclusively with female eggs, while male eggs are sorted out. The trays containing only female eggs are subsequently returned to the incubator.

The entire process, from start to finish, takes only 25 minutes.

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district – where the work was conducted. If the work was conducted outside of the US, please list the city and country.
Hatchery Het Anker, Ochten, the Netherlands; In Ovo lab BioPartner, Leiden, the Netherlands; Hatchery Vepymo, Poppel, Belgium
- 1.2. How many new jobs were created because of this grant funding?
2
- 1.3. How many jobs were maintained because of this grant funding?
9
- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.
No
- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.
No

2. Scientific Report

2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Public abstract

In Ovo was founded on the premise of finding a solution to the long-standing practice of culling male chicks the day they hatch – because they do not lay eggs. Culling of male chicks is not only unethical, but also extremely wasteful. Furthermore, male eggs occupy unnecessary space in hatcheries, meaning energy and money can be saved by eliminating them early from the incubation process.

After many years of research and development, In Ovo has built a market-ready solution for early gender screening – Ella. Ella is a machine that determines the sex of an egg instead of a chick, ensuring only females are hatched, thereby preventing the culling of male chicks. The male eggs are repurposed and valorized in the production of sustainable egg protein used in pet food. Hatching only female chicks frees up valuable setter and hatcher capacity, reducing the costs of the breeding process and lowering the

environmental footprint. As a fully automated, inline solution, Ella can determine the sex of an egg early in the incubation period. When developing this machine and technology, we worked closely with the sector so that hatcheries could easily integrate with Ella. Ella as such provides a solution to the culling of male chicks which can be implemented in any hatchery.

2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

Applicant:

Each year 6.5 billion female layer chicks hatch worldwide to become layer hens. At the same time the same number of male chicks are culled directly after hatch because they cannot lay eggs. These male chicks are deemed unsuitable for consumption in most western markets due to their poor growth and meat quality, so rearing those males is economically unfavorable.

However, male chick culling does not only represent an economic and environmental loss. Both farmers and consumers are starting to become increasingly concerned about animal welfare issues in farming. Consumers are increasingly tuned into animal welfare and environmental concerns when making decisions about what to buy.

Of the 6.5 billion chicks that hatch each year, 300 million are hatching in the European Union. In Europe the increasing pressure from animal welfare groups and the public has led to legislation addressing the ethical issues surrounding chick culling. In 2022 Germany became the first country to ban chick culling, by requiring hatcheries to use more humane alternatives like in-ovo sexing, dual-purpose chickens, or raising the males. From January 2024, an addition was made to the German legislation stating that in ovo sexing has to be performed before the 13th day of incubation. In 2023 France became the second country to ban chick culling and in ovo sexing must be performed before the 15th day of incubation. Italy will ban chick culling from 2027 onwards. With the success of Germany and France and the planned ban in Italy, there's growing momentum for the European Union to adopt comprehensive regulations that eliminate chick culling across all member states. Also in the US consumers are more than ever well-informed about animal welfare issues and this awareness has led to increased concern about the treatment of animals in industrial farming practices ([2023 Industrial Ag Survey Results Report](#)). This is well illustrated by the fact that early December 2024 the first in-ovo gendered eggs hatched in the U.S. ([NestFresh celebrates first hatch of in-ovo sexed chicks in the US](#))

Just like in Europe, pressure from US consumers and animal welfare groups drive the future banning of chick culling. According to figures from the International Egg Committee, approximately 308-million-layers are used in egg production in the United States in 2023 and 278 million are replaced yearly. This indicates another 278 million male chicks are

killed in the United States every year the day they hatch. Application of our in-ovo sexing technology, called Ella, offers a scalable solution to the worldwide chick culling problem. Currently two Ella machines are placed in a hatchery in the Netherlands and one in a Belgian hatchery. Our machines consist of several individual modules that can easily be transported and implemented in any hatchery in the US. Furthermore, our gendering solution is readily scalable to 15.000 or 20.000 eggs / hour. Consequently, Ella is ready for use in the U.S. agriculture and food systems. Applying this technology for early gender screening will not only address an ethical question but will also decrease the carbon footprint of hatcheries by reducing the incubation period of male eggs. We are currently testing from 9 until day 11, so on average on day 10. Removing all unfertilized eggs, early deaths and males (in total 53.5% of incubated eggs) on day 10 of the 18-day incubation process results in 24% of setter capacity reduction. Hatching capacity is reduced by 50% for the 3-day hatching period, resulting in a total energy reduction of 28%. In the Netherlands, we have ensured that the male eggs taken out of the incubation process are indicated as a category 3 waste stream and are allowed to be processed as an ingredient for pet food, a similar approach could be taken for the U.S.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- b) intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Applicant:

As explained above, pressure from consumers as well as animal welfare groups and changing legislation are key drivers in banning chick culling, calling the need for gendering solutions. Using in ovo sexing addresses these concerns and adds value to these chicks resulting in a positive business case for hatcheries. An [Agfunder](#) article on the introduction of cull free eggs by Nestfresh mentions that the extra value of a cull-free chick is \$1.50. As Ella works on all breeds and colors, it offers a solution to the whole US market.

Next to the added value, economic drivers for hatcheries are also an important factor in pushing in ovo gendering technology. As indicated above, approximately 278 million male chicks hatch every year in the U.S., which is not only unethical but also extremely wasteful. Our gendering solution provides a 28% reduction in setter energy capacity needed, as we take out all infertile and male eggs (combined: 50%) at on average day 10 of the 18-day incubation period. This reduces both the costs of the breeding process and lowers the environmental footprint.

Another cost saving can be found in manual sexing of chicks. We’re currently operating at an accuracy of 98.5% meaning almost all males are taken out of the incubation process at day 10, don’t hatch and don’t have to be manually sexed. When we produce consistently above 99% accuracy, the benchmark of manual sexing is reached, and it will no longer be necessary to do all together. The <1% male chicks will be reared alongside the females. We aim to reach the 99% accuracy this year. Bringing Ella to the US market will require additional adjustments to the current system to comply with US rules and regulations, as well as to scale the technology further to match with the needs of US poultry players. We are currently working on these further improvements and developments, which we aim to finish this year. As we have not yet placed our Ella in the U.S. market, we can only envisage the potential on how our gendering solution will impact U.S. agriculture.

[illegible]



2.5 Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

Applicant:

On May 24th, 2024, FFAR / Egg-Tech Prize communicated the obsolescence of the previous day criteria and provided new day criteria that the technology must function on or before day 11 of incubation. Consequently, we have adjusted this Milestone in the Increased gender accuracy section (see 2.6)

2.6 What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis on reporting in this section should focus on accomplishments. In response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Goal: Increased gender accuracy

We have reached the goal of a stable 98.5% gender accuracy. To reach this goal we have implemented several changes with respect to the sample workflow as well as to our gender screening algorithm, which are listed below.

Milestone 1: Implementation of automated plate handling.

Human interference with sample plate handling has been fully eliminated in the workflow. We have installed a conveyor belt, a plate robot, and a liquid handler taking care of all sample handling steps. Within Ella, samples are taken from individual eggs and samples are subsequently deposited in a deep well sample plate. Deep well plates originating from the sampler are transported to the analysis room by a conveyor belt. A robot arm picks up the plate from the belt and places it in the shuttle of the liquid handler. In the liquid handler samples are transferred into a new, specific ECHO-MS well plate. In the liquid handler several steps have been optimized which have resulted in a 25% reduction in handling time. After sample preparation, the robot arm picks up the Echo plate and places into the Echo MS device. During mass spectrometric analysis the original sample plate is stored in a well plate hotel. Here plates are saved until analysis, and gendering has been completed successfully to ensure the proper analysis of all samples taken. Finally, both deep-well and Echo plates are automatically discarded after analysis.



Milestone 2: Improved gender accuracy

Next to the implementation of the liquid handling systems, which has greatly enhanced reproducibility, we have further improved both the sampling as well as the analysis process. Consequently, we have reached a stable 98.5%, and often $\geq 99\%$ gender accuracy as stated in the submitted research proposal. This important milestone was achieved by 1) focusing on enhancing sample quality and 2) by improvement of biomarker accuracy. Below, each of these approaches is explained in more detail.

First, we have found that sample quality has a major impact on gender accuracy. To improve our gender accuracy and monitor performance during production we have implemented qPCR-based gender screening to cross-validate our gendering results and predict gender accuracy prior to hatch. Based on the results of this screening we found that Ella sampler performance is a key factor in determining sample quality. We have subsequently made additional modifications to the hardware as well as software in the Ella sampler. We found that a differential pressure approach (DPA) to take samples and transport them to the well plate is superior to a system which uses a peristaltic pump system. We therefore equipped the Ella sampler with a DPA system.

Second, we have continued our work on finding additional biomarkers that could be combined with our current biomarker to further improve gender accuracy. We have screened approximately 250 transitions for predictive value in gender screening. Based on these experiments we are testing the addition of extra biomarkers to our screening method which provides additional gender data.

Finally, we have overhauled our gendering algorithm to consider fluctuations in sample quality due to lower egg quality, small differences in egg size and incubation times. Taken together these modifications have led to a tremendous increase in gendering accuracy. Consequently, we currently run at a stable 98.5% gendering accuracy at hatchery 'het Anker'. We have already produced batches at $>99\%$ gender accuracy and expect with the above-mentioned developments to achieve a situation in 2025 where we consistently produce with an accuracy above 99%. Currently we have chosen to run at a stable 98.5% and use the improved gender separation to further improve our test ratio. The test ratio is an important parameter too, as it indicates which number of input eggs are needed to create one healthy female layer chick. In our Hatchery 'het Anker' they have a normal egg ratio of 2.5, which means on average 2.5 eggs are needed to create a healthy female. This number is based on the fact that in each batch a number of eggs are unfertilized, and a number of eggs do not hatch due to developmental defects (these two add up to 20% of incubated eggs). From the hatching eggs 50% is male and 50% is female. Starting from a gender accuracy of 96% in 2022 we have now a reproducible gender accuracy of over 98.5% and often batches reach a gender accuracy above 99% with a stable egg ratio of 2.7, which is approaching the hatcheries egg ratio for normal batches without in ovo sexing.

Milestone 3: Accurate and Robust sampling at 8 DOE

On May 24th, 2024, FFAR / Egg-Tech Prize communicated the obsolescence of the previous day criteria and provided a new day criteria that the technology must function on or before day 11 of incubation. Consequently, the urgency for improving screening solutions at earlier ages has decreased. However, our company perspective remains that we believe the value of our solution is maximized by screening as early as possible. As we know our biomarker is present on day 6 of incubation, screening on this day is our target. We're approaching day 6 gradually and are currently optimizing day 8 screening. The following changes we have made to our system have contributed to improved sampling at 8 DOE:

- Improved Vision algorithm for better allocation of the allantoic fluid before sampling
- A custom "recipe" for applying the right needle depth per egg, taking into account next to day of testing, variables such as Parent Stock age, egg size and breed.

To improve accurate sampling and to become more robust in our sampling we have introduced monitoring systems that constantly display the performance of the machine with respect to sample success rate and sample quality. These dashboards allow technical operators and engineers to adjust the sampler immediately when performance drops. In addition, we have implemented standard operating procedures; these have increased the robustness of the machine tremendously. Furthermore, training procedures for operators have enhanced the reliability of the sampler even further. Also as explained under milestone 2 we have made several changes to the sampler including the installation of the DPA system. This has led to major improvements in sample quality which immediately resulted in an increase in gender accuracy. In addition, the DPA system requires significantly less maintenance, further increasing the robustness of the system.

As mentioned earlier, we have multiple, additional, biomarker candidates that can be used to increase accuracy. These additional biomarkers have been shown to be present on earlier days as well and can be used for increased accuracy at, especially, early day samples. Our current biomarker, for example, is present as early as day 6, however it does not have a 99% predictive value at that egg age. Thanks to the nature of our screening technology, mass spectrometry, we can measure more than one biomarker per analysis, making it possible to continuously improve the screening method without losing speed and throughput. For gender screening at younger egg age, an additional marker can be added to our current marker screen to bring back the gender accuracy to 99% also at this age. At the moment we are test-running 3 of these additional markers for their predictive value and we anticipate finishing the analysis within the next 3 months. This versatile setup opens a path for additional applications with this platform, for example a biomarker that would predict chick quality or indicate developmental defects.

Goal: Reduced sample processing time

We have reached the goal and reduced sample processing time by 20%. To reach this goal we have implemented several changes with respect to the biomarker analysis workflow, which are listed below.

Milestone 4: Shorten the MS analysis time

We have established a 20% reduction in analysis time by automating sample handling, sample preparation and analysis. We have updated the software to allow batch dependent analysis, essentially meaning that only successfully sampled eggs will be analyzed, shortening the analysis time. We have also recently implemented a shorter inter sample delay time, this means that the time between subsequent ejections from the Echo has been reduced. This has shortened the analysis time tremendously and sampler and ECHO-MS analysis time are now synchronized in such a way that the sampler, sorter and analysis are working in parallel. We have already identified new modifications to the workflow which can speed up the process even further. We are aiming to achieve a 1 second analysis time per sample for the entire process of measurement and quantification. In collaboration with SCIEX, we are working on a new version of the deconvolution software. This can greatly enhance analysis speed as peaks from subsequent samples can overlap during analysis and are only deconvoluted by the analysis software. Again, we have not yet implemented this, but we are investigating in a research setting whether this still results in the desired gender accuracy. Finally, we are in the process of testing several new buffer compositions and MS settings for analysis to increase peak intensity and reduce peak broadening during the analysis. When peak width is reduced while maintaining good sensitivity and selectivity, more samples can be analyzed in the same amount of time.

Milestone 5: Multiplexing samples by modification

To find alternatives to reduce the analysis time per sample we have investigated the possibilities of multiplexing samples. This means that half of the samples are modified by a chemical reaction while the other half of the samples remain unmodified. Afterwards one modified and one unmodified sample is subsequently combined. In our mass spectrometric analysis, we have the possibility to analyze and quantify both the modified and unmodified sample in a single measurement, because the modified biomarker has a different mass. Potentially this approach could lead to a reduction in analysis time of 50% (not considering the time it takes to perform the modification reaction).

In collaboration with two external companies, we have investigated the possibility of multiplexing reagents. Multiplexing has several requirements that limit the possibility of modifying reagents and modifications sites within the current marker. At this stage of the process, we have indications that chemical modification is possible at a relatively high reaction rate, meaning the reaction is complete within a few minutes. However, modification

of our biomarker changes its physicochemical properties. In short this means that the characteristics of the modified marker are different from the unmodified marker. This means that the quantitative information that is extracted from these measurements cannot directly be compared. Consequently, combining samples is at this stage not possible. A solution to this would be the incorporation of stable isotopes in the chemical reagent. When using stable isotope labelled chemical modification reagents, all properties, but mass stay the same. In that case we could even expand the multiplexing to 3 or 4 levels. The downside of using stable isotopes is the high costs of such reagents. It is important to note that multiplexing is only beneficial if the reaction time is fast (below the time required to run a single plate) and the costs are effective. Alternative solutions to reducing analysis time, such as mentioned under Milestone 4, are easier to implement and result in a shorter analysis time within a much shorter time frame. Consequently, we decided to focus on alternatives as mentioned in milestone 4, which indeed have resulted in the desired increase in throughput.

Goal: Reduced hatch impact

We have reduced the hatch impact to below 1.5%. To reach this goal we have implemented several changes to the sampler monitoring process and hygiene monitoring and hygiene protocols, which are listed below.

Milestone 6: Provide clear evidence for minimal hatchability impact, i.e. <1.5%

We have reached a minimal hatchability impact of below 1.5% with our Ella. Already in the first months of the funding period a < 1.5% hatch impact was reached a couple of times, however it was not very consistent. Therefore, a specialized lead animal scientist was hired to investigate the source of these fluctuations and coordinate the efforts in reducing hatch impact.

There are various insidious factors which may contribute to our overall hatch impact; the main one that In Ovo has been battling has been hygiene and bacterial contamination (see Milestone 7). To prevent contamination, eggs going into the system were already sanitized with UVC light, egg punches were heated for disinfection during the process, and needles were cleaned after each sample collection.

Based on the findings of our lead animal scientist, the following improvements were made to further improve hygiene:

- The addition of applying H₂O₂ to the eggs before the UVC treatment, which in combination with the UVC treatment eliminates >99% of pathogens.
- New protocols to monitor hygiene levels of the machine and the facility, as well as proper cleaning protocols. We ensure eggs do not cool down during the sampling process and limit the time eggs are outside the setter.
- Closely monitoring eggshell temperatures during the entire sampling and sorting process. Hatch impact tests were carried out to determine the detrimental effects of lower

eggshell temperatures on hatch impact to help guide our best practices.

- Closely monitoring shocks during the entire sampling and sorting process. Egg sensors are placed on trays in the setters and are monitored throughout the Ella process. Shocks, eggshell temperature, environmental temperature and humidity are monitored during this time to ensure the eggs are handled with care. We have implemented several modifications to the Ella sampler to reduce shocks and temperature fluctuations during the sampling process.

Also, the reduction of the amount of human handling steps has led to a consistent reduction in hatch impact. To put this into context, the hatchery where our two gendering machines are installed reports an annual average hatchability of 93% without using Ella, with a 7% unhatched rate. We benchmark our performance against this baseline. For example, over the past four weeks (weeks 42–46), the average hatchability of gendered eggs was 92.5%, with a standard deviation of 0.6%. This demonstrates that our system performs consistently, with only a -0.5% difference compared to the hatchery's annual average.

Hatchability

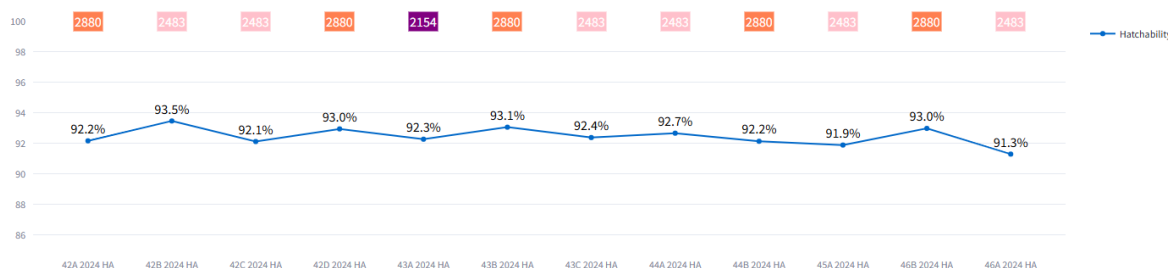


Figure 1. Hatchability after in ovo sexing with Ella in week 42-46, 2024

Milestone 7: No cross contamination of pathogens

We have eliminated the cross contamination of pathogens within the sampler entirely. First, eggs are sanitized with H_2O_2 prior to treatment with UVC, eliminating >99% of pathogens present at the eggshell.

Issues with cleaning blocks being blocked by debris caused by corrosion due to a cleaning product were contributing factors to the bacterial contamination we saw last year. Together with ensuring components are resistant to the cleaning agents and improvements in overall hygiene including hygiene awareness training of staff and third-party cleaning contractors, we have managed to remove the bacterial contamination portion from the hatch impact. Cross-contamination from egg to egg via punches or needles is ruled out as punches are heated for disinfection during the process, and needles are cleaned with disinfectants after each sample collection. To ensure these hygiene procedures work as intended we take regular swabs of needles and punches. In general, within our quality control we perform weekly swabs of machines and instruments that are in the Ella room, and which are directly or indirectly in contact with our eggs. Furthermore, our quality control comprises breakout analysis of unhatched eggs during hatches. When contamination is expected, swabs are sent out for analysis and determination of the pathogen involved.

Together with hatchery het Anker we have worked on improving the overall hygiene at hatchery, mainly leading to a massive reduction in fungal contamination.

Goal: Reduced human interference

We have reduced human interference by 50% during the sorting process.

Milestone 8: Automated sorting

Overall human interference has been reduced by 50% within the Ella system. We eliminated the manual candling and unfertile removal which was replaced by automated candling and flex picking. Unfertilized eggs are subsequently transported on conveyor belts and disposed. Automated sorting was implemented in the last year and in recent months we have implemented a faster sorter head. To reduce human interference even further, gendered males are sorted out of the trays immediately by our automated sorter, transported on conveyor belts and subsequently disposed.

2.7 Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

Applicant:

With every improvement project or modification to hardware or software we take a scientific approach such that we make sure that the results lead to a statistically significant result. To ensure a proper scientific approach we have implemented a test request system in which modification or tests within the Ella system are evaluated by experts coming from different areas within our company. Due to biological variability and different origin (multiplier variation) there is some natural variation in egg quality and consequently sample quality. We collect massive amounts of data from our production locations, allowing us to monitor performance over time. In addition, this data collection allows us to perform a meta-analysis of all datasets collected. This means we can retrospectively perform correlation analysis, using different input factors. For example, there is a clear correlation between parent flock age and egg quality, which becomes clearly visible when collecting larger datasets. In the future this meta-data could also be used as predictive models to deal better with variations in sample quality.

2.8 List key stakeholders that could be served by results of this project.

Applicant:

Almost the entire egg production chain are stakeholders in this project. This includes Hatcheries, Layer Farms, Egg packing industry, Supermarket chains and consumers. As explained in previous sections, in ovo sexing results in an increased value of the female chicks that hatch, which, combined with reduced energy and manual sexing costs, affects every step in the egg production chain.

Stakeholder	Benefit
Hatchery (Ella customer)	• Higher margin (added value, reduced energy & manual sexing costs) • In Ovo vaccination possibilities for layers • Complying with legislation, no chick culling after day 13 of incubation
Layer Farms	Added value to eggs, increased margin
Egg packing industry	Added value to eggs, increased margin
Supermarket	Added value to eggs, increased margin
Consumers	Customer satisfaction

2.9 How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Nothing to report

2.10 Describe challenges or delays encountered during the project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

Applicant:

As described in section 2.6 we have battled some pathogen contamination at the beginning of last year. Although this contamination did not originate from our sampler or gendering solution, it seriously impacted our hatch results. Together with hatchery het Anker we investigated the source of the contamination and found that the egg storage room of the hatchery itself was suffering from high fungal load. To eliminate the source of contamination we had to fog both the egg storage, setter halls and Ella room. Together with hatchery het Anker we have worked on improving the overall hygiene at the hatchery, leading to a massive reduction in fungal contamination. Due to these unforeseen circumstances, we reached our lower than 1.5% hatch impact later than originally anticipated. Also, as delineated in milestone 5, multiplexing experiments did not immediately lead to the anticipated results, however this did not lead to any delay in progress with respect to increasing speed of the analysis as alternative solutions were applied.

2.11 What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Applicant:

During this project several changes were made to the Ella sampler as well as to the workflow. We quickly realized that our operators needed training as well as standardized ways of working.

Therefore, we introduced a standardized way of working which we developed with an external company, called Training within Industry (TWI). A central role within TWI is provided by employees which are educated internally to become specialized trainers for new employees. We now have internal trainers for each discipline within production and the company. This internal training for new employees makes sure that all the operators have the same and adequate training for their tasks. Next to proper training we have developed and written standardized protocols that are available via Elicit Online. Elicit Online provides a knowledge structure which is based on experiential knowledge. Elicit Online helped to optimize our operational performance, by providing standardized ways of decision making and problem solving, based on experience from expert users. We also use Elicit to control the daily, weekly and monthly quality checks that need to be performed



within each discipline. Finally, to reduce the impact of human error we introduced company-wide hygiene and safety training.

2.12 Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

NA



3 Information Products

- 3.4 Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Applicant:

In the period 2022 – 2024 we have rolled out a PR and marketing strategy that was mainly focused on addressing the issue that the world is dealing with (the yearly culling of 6,5 billion-day-old chicks worldwide) and the fact that we (amongst others) are working on a solution. We frequently publish news items on this subject on LinkedIn and the collection can be found on our website:

<https://inovo.nl/news/>

From 2021 In Ovo is running in several hatcheries, which makes the message even stronger, since we are commercially operational. For the UK market, we also did a consumer research, which resulted in insights into the opinion of the UK consumer on the topic of chick culling and animal welfare. On top of that, we hired a researcher to investigate persistent (incorrect) stories on the chick culling industry. The results have been helpful in conversations with media and NGO's, but not resulted in media coverage yet. Furthermore, we engaged with several media outlets over time for opportunities to talk about our company, the issues we face and the solution we offer. This resulted in several radio appearances and multiple interviews with one of our founders, Wouter Bruins. He was also on stage at the yearly TNW Conference, a highly regarded international tech conference.

<https://www.bnr.nl/podcast/wereldspelers/10552785/de-nederlandse-boerensector-is-nog-groot-over-de-grens>

<https://www.bnr.nl/podcast/wereldspelers/10536884/problemen-voor-asml-met-china-miljarden-kuikens-redden>

- 3.2 Please provide a list of citations for the information products produced during the project.

NA

- 3.3 Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

NA

- 3.5 Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

NA

- 3.6 Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s). **NA**

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

- 3.7 Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

NA

- 3.8 Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

NA

4 Data Management

4.4 Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

Applicant:

We don't store our production data in a public repository as this data is only used for gender prediction. During this project no new research data has been generated. We do store all the data generated by Ella as well as all gendering data generated by our Echo-MS systems. In addition, we store all data related to instrument performance as well as batch related information (parent age, egg age, storage days, multiplier etc.) Accumulating these large datasets allows us to perform a correlation and meta-analysis of internal and external factors that affect or key performance indicators (KPIs), Sample Success rate, Gender accuracy, Hatch impact and test ratio. All information collected is represented in informative dashboards only open to company employees. Collecting and correlating this data is crucial for further development and improvement within the company. For example, there is a correlation between egg size and biomarker concentration which can negatively affect gender accuracy. From this observation we are currently estimating if we can correct for variations in egg size within our marker analysis. As another example, parent age and storage time are crucial for sample success rate as well as hatchability. By having this information beforehand, we can tune our instruments to better deal with such batches. All production and hatch data are also shared with the hatcheries via our online dashboard Conneggt.

4.5 If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

NA

4.6 Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

NA



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RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Simone Lemeer

PI Signature:

A handwritten signature in blue ink that reads "Slemeer".

Date: 13-01-25

Authorized Signing Official (ASO) Name: Tiemen Dalhuisen

ASO Signature:

A handwritten signature in black ink, consisting of a large, stylized loop followed by a horizontal line.

Date: 13-01-25